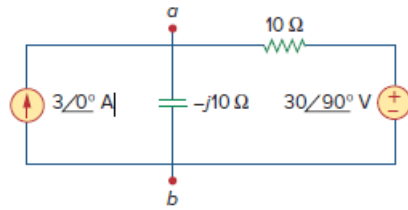


Problem

For the circuit depicted in Fig. 10.101, find the Thevenin equivalent circuit at terminals a - b .

Figure 10.101:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.101 in the textbook.

To calculate Thevenin impedance, open circuit the $3\angle 0^\circ$ A current source and short circuit the $30\angle 90^\circ$ V voltage source. Remove the $-j10\ \Omega$ impedance between the terminals a - b as shown in Figure 1.

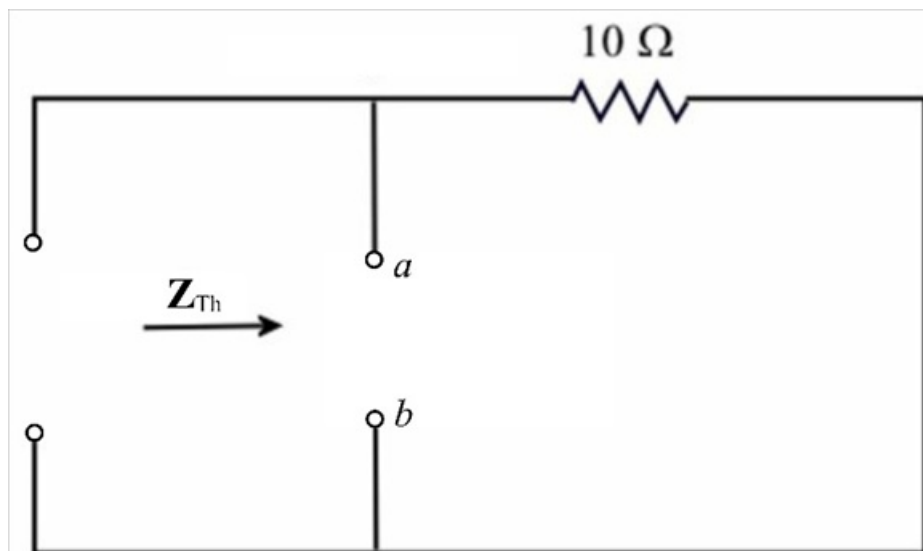


Figure 1

Step 2 of 4

The Thevenin impedance is the parallel combination of $10\ \Omega$ resistor and the open circuit (infinite impedance).

$$Z_{Th} = (10) \parallel (\infty)$$

$$\frac{1}{Z_{Th}} = \frac{1}{10} + \frac{1}{\infty}$$

$$\frac{1}{Z_{Th}} = \frac{1}{10}$$

$$Z_{Th} = 10\ \Omega$$

Step 3 of 4

Remove the $-j10\ \Omega$ impedance between the terminals a - b and calculate the Thevenin voltage V_{Th} , as shown in Figure 2.

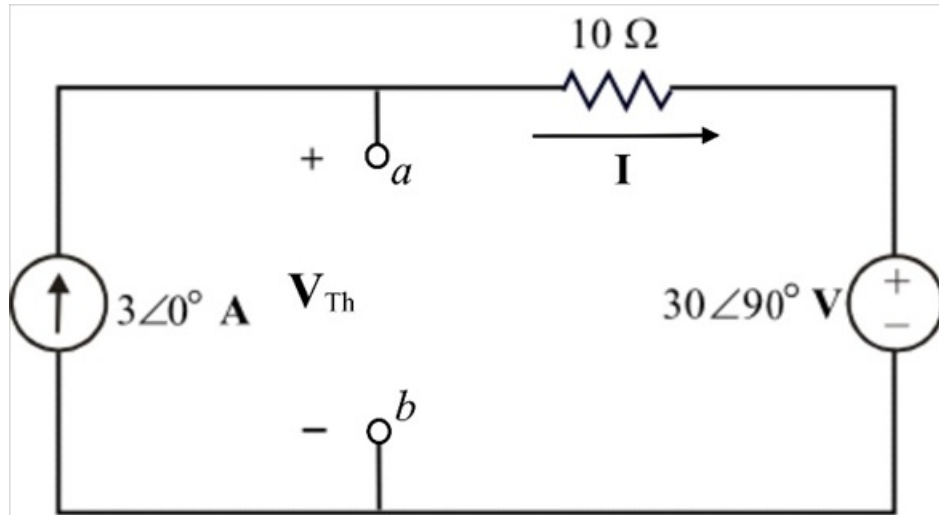


Figure 2

The current I is same as current source $3\angle 0^\circ \text{ A}$ since the terminal a - b is open.

$$I = 3\angle 0^\circ \text{ A}$$

Apply Kirchhoff's voltage law in the loop containing 10Ω resistor, $30\angle 90^\circ \text{ V}$ voltage source, and voltage V_{Th} .

$$-V_{Th} + 10I + 30\angle 90^\circ = 0$$

$$V_{Th} = 10I + 30\angle 90^\circ$$

Substitute $3\angle 0^\circ \text{ A}$ for I .

$$V_{Th} = 10(3\angle 0^\circ) + 30\angle 90^\circ$$

$$= 30\angle 0^\circ + 30\angle 90^\circ$$

$$= 30(\cos 0^\circ + j \sin 0^\circ) + 30(\cos 90^\circ + j \sin 90^\circ)$$

$$= 30(1) + 30(j)$$

$$= 30 + 30j$$

$$= \sqrt{30^2 + 30^2} \tan^{-1} \left(\frac{30}{30} \right)$$

$$= 42.43\angle 45^\circ \text{ V}$$

The Thevenin equivalent circuit is the Thevenin voltage source ($V_{Th} = 42.43\angle 45^\circ \text{ V}$) in series with the Thevenin impedance ($Z_{Th} = 10 \Omega$).

Draw the Thevenin equivalent circuit as shown in Figure 3.

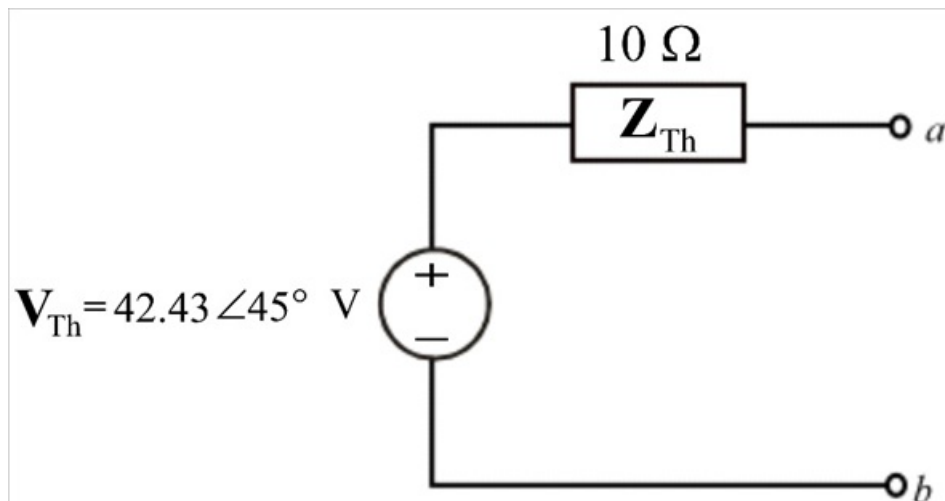


Figure 3